Anas Barakat

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RESEARCH PROFILE

Optimization and learning for strategic and sequential decision making:

- **Policy optimization beyond expected additive rewards:** general-utility RL and post-training.
- **Learning and optimization in structured games:** multi-agent RL, Markov games with general utilities, learning dynamics, and equilibrium learning.
- **Online learning beyond convexity under limited feedback:** online learning, bandit and preference feedback, nonconvex optimization, and hidden convexity.

ACADEMIC APPOINTMENTS

Singapore University of Technology and Design
Research Fellow — Oct. 2024-present
Hosts: Prof. Georgios Piliouras and Prof. Antonios Varvitsiotis

ETH Zurich, Department of Computer Science
Foundations of Data Science Postdoctoral Fellow — Feb. 2022 - Aug. 2024
Host: Prof. Niao He

EDUCATION

Institut Polytechnique de Paris, Télécom Paris, Paris, France
Ph.D. in Applied Mathematics and Computer Science — 2018 - 2021
Advisors: Prof. Pascal Bianchi and Prof. Walid Hachem

Université Paris Saclay, Paris, France
M.Sc. in Data Science (with highest honors) — 2017 - 2018

Télécom Paris, Paris, France
M.Sc. in Applied Mathematics and Computer Science — 2015 - 2018

SELECTED PUBLICATIONS

1. **Anas Barakat**, Souradip Chakraborty, Khushbu Pahwa, Amrit Singh Bedi. *Prompt Interference in LLM Post-Training: Why Pass@k Optimization Can Degrade Pass@1.* NeurIPS 2026.
2. **Anas Barakat**. *Learning in continuous games from pairwise preference feedback.* Preprint, 2026.
3. **Anas Barakat**, Andreas Kontogiannis, Vasilis Pollatos, Ioannis Panageas, Antonios Varvitsiotis. *Online learning on hidden-convex losses via algorithmic equivalence: optimal regret, geometric barrier, and bandit feedback.* NeurIPS 2026.
4. **Anas Barakat**, Souradip Chakraborty, Peihong Yu, Pratap Tokekar, Amrit Singh Bedi. *On the global optimality of policy gradient methods in general utility reinforcement learning.* NeurIPS 2025.
5. **Anas Barakat**, Ioannis Panageas, Antonios Varvitsiotis. *Convex Markov Games and beyond: new proof of existence, characterization and learning algorithms for Nash equilibria.* AISTATS 2026.
6. John Lazarsfeld, **Anas Barakat**, Georgios Piliouras, Antonios Varvitsiotis, Andre Wibisono. *When and why is optimistic multiplicative weights slow? The geometry of energy dissipation.* NeurIPS 2026.
7. **Anas Barakat**, John Lazarsfeld, Georgios Piliouras, Antonios Varvitsiotis. *Online learning in stabilized linear dynamical games with adversarial disturbances.* NeurIPS 2026.

PUBLICATIONS & PREPRINTS

Names of student co-authors I have previously supervised and mentored are underlined.

Preprints

(1) **Anas Barakat**. *Learning in Continuous Games from Pairwise Preference Feedback.* Preprint 2026.

(2) John Lazarsfeld, **Anas Barakat**, Jacob Abernethy, Vidya Muthukumar, Georgios Piliouras, Andre Wibisono, Juba Ziani. *Linear Last-Iterate Convergence Rates for Optimistic Learning In Harmonic Games.* Preprint 2026.

Refereed Conference Publications

(C1) **Anas Barakat**, Souradip Chakraborty, Khushbu Pahwa, Amrit Singh Bedi. *Prompt interference in LLM post-training: Why pass@k optimization can degrade pass@1.* Advances in Neural Information Processing Systems (NeurIPS 2026).

(C2) **Anas Barakat**, Andreas Kontogiannis, Vasilis Pollatos, Ioannis Panageas, Antonios Varvitsiotis. *Online Learning on Hidden-Convex Losses via Algorithmic Equivalence: Optimal Regret, Geometric Barrier, and Bandit Feedback.* Advances in Neural Information Processing Systems (NeurIPS 2026).

(C3) John Lazarsfeld, **Anas Barakat**, Georgios Piliouras, Antonios Varvitsiotis, Andre Wibisono. *When and Why is Optimistic Multiplicative Weights Slow? The Geometry of Energy Dissipation.* Advances in Neural Information Processing Systems (NeurIPS 2026).

(C4) **Anas Barakat**, John Lazarsfeld, Georgios Piliouras, Antonios Varvitsiotis. *Online Learning in Stabilized Linear Dynamical Games with Adversarial Disturbances.* Advances in Neural Information Processing Systems (NeurIPS 2026).

(C5) **Anas Barakat**, Ioannis Panageas, Antonios Varvitsiotis. *Convex Markov Games and Beyond: New Proof of Existence, Characterization and Learning Algorithms for Nash Equilibria.* International Conference on Artificial Intelligence and Statistics (AISTATS 2026).

(C6) **Anas Barakat**, Souradip Chakraborty, Peihong Yu, Pratap Tokekar, Amrit Singh Bedi. *On the Global Optimality of Policy Gradient Methods in General Utility Reinforcement Learning.* Advances in Neural Information Processing Systems (NeurIPS 2025).

(C7) Kimon Protopapas, **Anas Barakat**. *Policy Mirror Descent with Lookahead.* Advances in Neural Information Processing Systems (NeurIPS 2024).

(C8) Pragnya Alatur, **Anas Barakat***, Niao He. *Independent Policy Mirror Descent for Markov Potential Games: Scaling to Large Number of Players.* IEEE Conference on Decision and Control (CDC 2024). *Corresponding author.

(C9) Philip Jordan, **Anas Barakat**, Niao He. *Independent Learning in Constrained Markov Potential Games.* International Conference on Artificial Intelligence and Statistics (AISTATS 2024).

(C10) Jiduan Wu, **Anas Barakat**, Ilyas Fatkhullin, Niao He. *Learning Zero-Sum Linear Quadratic Games with Improved Sample Complexity.* IEEE Conference on Decision and Control (CDC 2023).

(C11) **Anas Barakat**, Ilyas Fatkhullin, Niao He. *Reinforcement Learning with General Utilities: Simpler Variance Reduction and Large State-Action Space.* International Conference on Machine Learning (ICML 2023).

(C12) Ilyas Fatkhullin, **Anas Barakat**, Anastasia Kireeva, Niao He. *Stochastic Policy Gradient Methods: Improved Sample Complexity for Fisher-non-degenerate Policies.* International Conference on Machine Learning (ICML 2023).

(C13) **Anas Barakat**, Pascal Bianchi, Julien Lehmann. *Analysis of a Target-Based Actor-Critic Algorithm with Linear Function Approximation.* International Conference on Artificial Intelligence and Statistics (AISTATS 2022).

(C14) **Anas Barakat**, Pascal Bianchi. *Convergence Rates of a Momentum Algorithm with Bounded Adaptive Step Size for Nonconvex Optimization.* Asian Conference on Machine Learning (2020).

Refereed Journal Publications

(J1) Olivier Lepel, **Anas Barakat**. *Policy Gradients for Cumulative Prospect Theory in Reinforcement Learning.* Transactions on Machine Learning Research, 2026.

(J2) **Anas Barakat**, Wayne Lin, John Lazarsfeld, Antonios Varvitsiotis. *Optimistic Online Learning in Symmetric Cone Games.* Transactions on Machine Learning Research, 2026.

(J3) Jiduan Wu, **Anas Barakat**, Ilyas Fatkhullin, Niao He. *Learning Zero-Sum Linear Quadratic Games with Improved Sample Complexity and Last Iterate Convergence.* SIAM Journal on Control and Optimization, 2025.

(J4) **Anas Barakat**, Pascal Bianchi, Walid Hachem, Sholom Schechtman. *Stochastic optimization with momentum: convergence, fluctuations, and traps avoidance.* Electronic Journal of Statistics 15 (2), 3892-3947, 2021.

(J5) **Anas Barakat**, Pascal Bianchi. *Convergence and Dynamical Behavior of the Adam Algorithm for Non-Convex Stochastic Optimization.* SIAM Journal on Optimization 31 (1), 244-274, 2021.

AWARDS AND SCHOLARSHIPS

- ETH Zurich Foundations of Data Science Postdoctoral Fellowship 2022–2024
- DAAD Postdoctoral Networking Tour in AI (Postdoc-NeT-AI) Fellow Nov. 2023
- Dodu Prize, French Society of Applied and Industrial Mathematics (SMAI) Sep. 2021
  Best communication by a young researcher at the Optimization and Decision annual meeting
- Mines-Télécom Institute (IMT) Ph.D. Scholarship 2018–2021
- Moroccan Government Merit Scholarship 2015–2018
- Agency for French Education Abroad (AEFE) "Excellence-Major" Scholarship 2013–2015

STUDENT RESEARCH MENTORING

Led the supervision of six master's/semester research projects; five led to refereed publications.

- Olivier Lepel — Master's thesis, Mar.–Sep. 2024. *Beyond Expected Returns: A Policy Gradient Algorithm for Cumulative Prospect Theoretic Reinforcement Learning.* TMLR 2026.
- Kimon Protopapas — Master's semester project, Sep. 2023–Jan. 2024. *Policy Mirror Descent with Lookahead.* NeurIPS 2024.
- Philip Jordan — Master's thesis, May–Dec. 2023. *Independent Learning in Constrained Markov Potential Games.* AISTATS 2024.
- Jiduan Wu — Master's thesis, Oct. 2022–Mar. 2023. *Learning Zero-Sum Linear Quadratic Games with Improved Sample Complexity and Last-Iterate Convergence.* IEEE CDC 2023.
- Harish Rajagopal — Master's thesis, Mar.–Sep. 2022. *Multistage Step Size Scheduling for Minimax Problems.*
- Julien Lehmann — Master's thesis, May–Oct. 2021. *Analysis of a Target-Based Actor-Critic Algorithm with Linear Function Approximation.* AISTATS 2022.

SELECTED INVITED TALKS AND SEMINARS

- Invited talk, 5th Symposium on Machine Learning and Dynamical Systems — 10 Feb. 2026
  Kyoto University, Kyoto, Japan
- Invited talk, ICCOPT 2025, University of Southern California, Los Angeles, USA — 24 Jul. 2025
- Invited talk, Learning Theory and Applications Workshop — 29 Apr. 2025
  College of Computing and Data Science, Nanyang Technological University, Singapore
  Host: Themis Gouleakis.
- Online invited talk, Finance and RL Talks, FinRL core team & Melwy open AI lab — 21 Apr. 2025
  Host: Mostapha Benhenda.
- Invited talk, Fourth Symposium on Machine Learning and Dynamical Systems — 11 Jul. 2024
  Fields Institute, Toronto, Canada.
- Invited talk, ICCOPT 2022, Lehigh University, Bethlehem, USA — 28 Jul. 2022
- Invited talk (online), 14th CMStatistics International Conference — 20 Dec. 2021
  Session: "Dynamical systems in machine learning" organized by Anna Korba
  King's College, London, United Kingdom
- Invited seminar, *Image, Optimization and Probabilities* research group — 15 Oct. 2020
  Bordeaux Institute of Mathematics (IMB), Bordeaux, France
- 2nd Symposium on Machine Learning and Dynamical Systems — 21 Sep. 2020
  Fields Institute for Research in Mathematical Sciences, online

TEACHING

- **Singapore University of Technology and Design**, Singapore
  - Co-designed and co-taught 'Online Learning and Learning in Games' — Fall 2025
    [course website]
- **ETH Zurich**, Zurich, Switzerland
  - Head Teaching Assistant for 'Foundations of Reinforcement Learning' — Spring 2023
  - Instructor for two lectures in 'Optimization for Data Science' — Spring 2023
  - Instructor for a lecture on 'Value-Based Methods' in 'Foundations of RL' — Spring 2024
  - Coordinator for the seminar course 'Advanced Topics in Machine Learning' — Fall 2022, 2023
  - Teaching Assistant for 'Optimization for Data Science' and 'Foundations of RL' — Spring 2024
- **Télécom Paris**, Paris, France
  - Teaching Assistant for 'Optimization for Machine Learning' — Fall 2018, 2019, 2020
  - Instructor for discrete Markov chains in 'Probabilities' — Fall 2018, 2019
  - Teaching Assistant for 'Probabilities', 'Statistics', 'Machine Learning' — Fall 2018, 2019
- **École Polytechnique**, Paris, France
  - Tutor: Design and Analysis of Algorithms; Advanced Programming ($\sim$ 20 hours) — Spring 2017

ACADEMIC SERVICE

- **Area Chair:** ICLR 2027.
- **Reviewer distinctions:** NeurIPS Top Reviewer 2026; ICML Gold Reviewer / Top 25% 2026; AISTATS Top 10% Reviewer 2022.
- **Conference reviewing:** NeurIPS (2022, 2023, 2025, 2026), ICML (2024, 2025, 2026), ICLR (2025-2026), AISTATS (2022, 2024, 2026), EC 2026.
- **Journal reviewing:** Journal of Machine Learning Research (JMLR), Transactions on Machine Learning Research (TMLR), Mathematical Programming, SIAM Journal on Optimization (SIOPT), Journal of Optimization Theory and Applications (JOTA), IEEE Transactions on Automatic Control (TACON), Stochastic Systems, Systems & Control Letters, Mathematics of Control, Signals, and Systems (MCSS), IEEE Control Systems Letters (IEEE L-CSS).
- **Organization of workshops, summer schools and conferences:**
  - Co-organizer, Multi-Agent RL EPFL–ETHZ Summer School 2024; co-developed successful CHF 20,000 funding proposal.
  - ICCOPT 2022, session organizer and chair: *Policy Gradient and Actor-Critic Methods: Theoretical Analysis and New Opportunities*, in the Optimization for Data Science and ML cluster.
  - Workshop organizer: RL workshop for industrial partners (Airbus Defence & Space, Engie, Idemia, Safran and Valeo) of the Data Science & AI for Digitalized Industry & Services research and teaching chair.